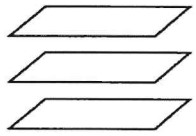
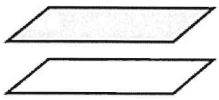
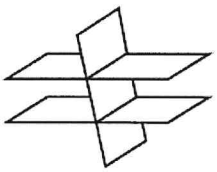
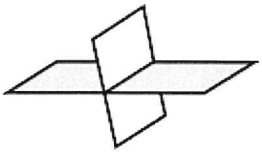
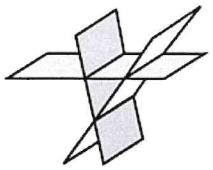
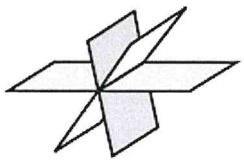
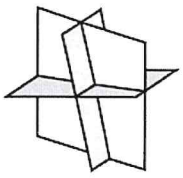
Intersection of Three Planes

Diagram	Description of Planes	Normal Vectors Comparison	# of Solution
	3 parallel planes	$\vec{n_1} \parallel \vec{n_2} \parallel \vec{n_3}$ (all parallel)	$\emptyset$
	2 coincident planes the other parallel	$\vec{n_1} \parallel \vec{n_2} \parallel \vec{n_3}$ (all parallel)	$\emptyset$
	3 coincident planes	$\vec{n_1} \parallel \vec{n_2} \parallel \vec{n_3}$ (all parallel)	Infinite # (plane)
	2 parallel and distinct planes the other crossing	$\vec{n_1} \parallel \vec{n_2}$	$\emptyset$
	2 planes coincident the other crossing	$\vec{n_1} \parallel \vec{n_2}$	Infinite # (Line)
	3 non-parallel planes but no intersection	normals are coplanar. $\vec{n_1} \cdot (\vec{n_2} \times \vec{n_3}) = 0$	$\emptyset$
	3 planes all meet at one line	normals are coplanar $\vec{n_1} \cdot (\vec{n_2} \times \vec{n_3}) = 0$	Infinite # (Line)
	3 planes intersect at one point	normals not parallel and not coplanar	1 (point)

Example 1: Determine the intersection of the planes → and classify the planes.

$$3x+2y+4z=9$$
$$x-y+3z=-2$$
$$x+4y-3z=14$$

$\vec{n_1}=(3,2,4)$, $\vec{n_2}=(1,-1,3)$, $\vec{n_3}=(1,4,-3)$ } not //

$\vec{n_1}\cdot(\vec{n_2}\times\vec{n_3})=5\neq 0$

$\therefore$ non-coplanar

$\therefore$ 1 p.o.i.

$$\left[\begin{array}{ccc|c} 3 & 2 & 4 & 9 \\ 1 & -1 & 3 & -2 \\ 1 & 4 & -3 & 14 \end{array}\right] \rightarrow \begin{matrix} R2 \\ R1 \\ \\ \text{eliminate} \end{matrix}\left[\begin{array}{ccc|c} 1 & -1 & 3 & -2 \\ 3 & 2 & 4 & 9 \\ 1 & 4 & -3 & 14 \end{array}\right] \rightarrow \begin{matrix} \\ 3R1-R2 \\ R1-R3 \\ \text{eliminate} \end{matrix}\left[\begin{array}{ccc|c} 1 & -1 & 3 & -2 \\ 0 & -5 & 5 & -15 \\ 0 & -5 & 6 & -16 \end{array}\right]$$

$$\begin{matrix} \\ -\frac{1}{5}R2 \\ R3-R2 \end{matrix}\left[\begin{array}{ccc|c} 1 & -1 & 3 & -2 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \rightarrow \begin{matrix} R1+R2 \\ \\ \\ \end{matrix}\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -1 \end{array}\right] \rightarrow \begin{matrix} R1-2R3 \\ R2+R3 \\ \\ \end{matrix}\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

↑ reduced Row-Echelon Form or Gauss-Jordan Form.

$\therefore (x,y,z)=(3,2,-1)$

$\therefore$ 3 non-parallel planes intersecting at one point.

Example 2: Determine the intersection of the planes → and classify the planes

$$x+y+2z=-2$$
$$3x-y+14z=6$$
$$x+2y=-5$$

$\vec{n_1}=(1,1,2)$, $\vec{n_2}=(3,-1,14)$, $\vec{n_3}=(1,2,0)$ } not //.

$\vec{n_1}\cdot(\vec{n_2}\times\vec{n_3})=0$.

$\therefore$ coplanar.

$\therefore$ no solution or a line.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & -2 \\ 3 & -1 & 14 & 6 \\ 1 & 2 & 0 & -5 \end{array}\right] \rightarrow \begin{matrix} \\ 3R1-R2 \\ R1-R3 \end{matrix}\left[\begin{array}{ccc|c} 1 & 1 & 2 & -2 \\ 0 & 4 & -8 & -12 \\ 0 & -1 & 2 & 3 \end{array}\right] \rightarrow \begin{matrix} \frac{1}{4}R2 \\ \\ \\ \end{matrix}\left[\begin{array}{ccc|c} 1 & 1 & 2 & -2 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & 2 & 3 \end{array}\right]$$

eliminate.

$$\begin{matrix} \\ R2+R3 \\ \\ \end{matrix}\left[\begin{array}{ccc|c} 1 & 1 & 2 & -2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightarrow \begin{matrix} R1-R2 \\ \\ \\ \end{matrix}\left[\begin{array}{ccc|c} 1 & 0 & 4 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightarrow \text{Let } z=t.$$

eliminate

R2: $y-2z=-3$, $y=2t-3$

R1: $x+4z=1$, $x=-4t+1$

$\therefore (x,y,z)=(1-4t,-3+2t,t)$ ← line of intersection

$=(1,-3,0)+t(-4,2,1)$, $t\in\mathbb{R}$.